

Tingjia Xu

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PROFILE

Research Interests: • Active Mobility • Built Environment & Travel Behavior • Healthy Cities • Urban Design

Selected Highlights: • World Bank Short-Term Consultant • Route Choice Analysis based on 3M+ Cycling Trajectories
• Project Lead for a Healthy Community Regeneration Handbook

EDUCATION

M.Eng. in Urban Planning

Sep. 2023–Jun. 2026

Tongji University | GPA: 4.54/5.00 (Top 2%) | Healthy City WLAN Lab | Advisor: Prof. Lan Wang

- China National Scholarship (2024, ranked 1st in cohort)
- Shanghai Outstanding Graduate Award (2026); Tongji Outstanding Student (two consecutive years)

B.Eng. in Urban and Rural Planning

Sep. 2018–Jun. 2023

Tongji University | GPA: 4.40/5.00 (Top 3%)

- Tongji Outstanding Undergraduate Thesis
- First-Class Scholarship; College Dean's Scholarship (highest college-level award)

CONSULTING AND INTERNSHIPS

World Bank Group, Office of the Chief Economist for Infrastructure (Washington, DC)

May 2026–Present

Short-Term Consultant | China (remote) | Project Lead: Yue Li, Senior Economist

- Developing a geospatial database of 300+ large-scale urban projects in China, including geocoding and performance indicators; processing project data for Indonesia.

Shanghai Tongji Urban Planning & Design Institute | Intern, Shanghai

Sep. 2023–Dec. 2025

- Conducted fieldwork, GIS analysis, and urban design across 4 planning projects (e.g., 15-minute community life-circle planning).

Asian Development Bank (ADB)–Tongji Urban Knowledge Hub | Intern, Shanghai

Sep. 2023–Jun. 2024

- Synthesized international cases into urban development reports (e.g., a report on healthy community regeneration).

COMPETITION AWARDS

Selected awards in planning, design & innovation (**1 international, 4 national, 3 provincial**):

2022 WUPEN iCity Urban Sustainability Report Competition	International	Honorable Mention
2024 Future Planners' Land Planning Design Competition — Graduate Division	National	First Prize
2024 "Beautiful China" Innovative Design Competition	National	First Prize
2023 Future Planners' Land Planning Design Competition — Undergraduate Division	National	First Prize
2024 China International College Students' Innovation Competition	National	Bronze Award
2024 Shanghai "Huichuang Youth" Urban Design Competition	Provincial	First Prize
2021 Shenchengli Micro-Regeneration Competition	Provincial	First Prize
2024 China International College Students' Innovation Competition – Shanghai	Provincial	Silver Award

RESEARCH PROJECTS

➤ Urban Mobility & Built Environment

The Impact of Built Environment on Cycling Route Choice

Sep. 2023–Jun. 2026

Master's Thesis | Advisor: Prof. Lan Wang

- Map-matched 3M+ dockless bike-sharing trajectories; extracted streetscape features using semantic segmentation.
- Estimated Path-Size Logit models to examine time-of-day variation in route preferences.
- Two first-author manuscripts (one under review; one in final revision for submission)

Space Syntax Analysis of Metro-Station Underground Space

Oct. 2024–Jan. 2025

Graduate Course Project | Team Lead | Advisor: Prof. Yu Ye | ACSP 2025

- Used Space Syntax to assess accessibility and intelligibility at Jing'an Temple Station.
- Applied visibility graph analysis and agent-based movement simulation to inform retail placement.

Streetscape Planning for Thermal Environment and Cycling Vitality

Jun. 2025–Jun. 2026

National Key R&D Program of China | Graduate Researcher | Advisors: Prof. Lan Wang; Asst. Prof. Lu Shan

- Built a street-segment dataset integrating cycling-volume, thermal, and streetscape attributes.
- Reviewed streetscape effects on urban heat and cycling vitality for a *Cities* manuscript.

➤ Urban Health & Livability

Health-Oriented Community Public-Space Regeneration

2022–2023

National Student Innovation Project | Project Lead | Advisors: Assoc. Prof. Yin Jie; Prof. Lan Wang

- Developed a four-pathway health framework and design strategies across five spatial domains.
- Led the team in producing two handbooks.

Heat-Adjusted Health Benefits of Cycling

Jun. 2025–Present

National Key R&D Program of China | Main Researcher | Advisors: Prof. Lan Wang; Asst. Prof. Wenyao Sun

- Co-developed the Heat-Exposure-Adjusted Cycling Health Benefit Index (HCBI) and estimated it at 100-m resolution across Shanghai.
- Co-authored an equal-contribution working paper and drafted a pending U.S. patent application.

Emergency Medical Services (EMS) Demand Forecasting and Resource Optimization

May 2026–Present

Student-led Research Commercialization | Main Researcher | Project Lead: Dr. Surong Zhang

- Developed a spatiotemporal emergency-call demand model; built the web platform and designed its UI/UX.
- Supported commercialization with a leading EMS dispatch provider serving 80% of the national market.

Social Media Sentiment Analysis for Perceived Livability

Apr. 2025–Nov. 2025

National Key R&D Program of China | Graduate Researcher | Advisors: Prof. Lan Wang; Assoc. Prof. Bochu Liu

- Synthesized assessment indicators across four livability domains (active travel, heat, noise, and light).
- Developed a four-dimensional livability lexicon to identify relevant social media posts.

PUBLICATIONS

Zhang, S., Wang, L. *, Shen, Y., Zhang, Y., Gao, Y., **Xu, T.**, & Zhang, Z. (2024). What causes Emergency Medical Services (EMS) delay? Unravel high-risk buildings using citywide ambulance trajectory data. *Habitat International*, 153, 103198. doi: 10.1016/j.habitatint.2024.103198

Xu, T., Wang, L. *, Zhang, S., & Sun, W. Research progress and future prospects on the impact of street space factors on cycling route choice. *Urban Development Studies* (under review; in Chinese).

Xu, T., Wang, L. *, Zhang, S., & Jia, Y. How do cyclists' street-environment preferences vary within a day? Evidence from dockless bike-sharing route choices in Shanghai (in final revision for submission).

Shan, L., **Xu, T.**, Zhou, Y., Zhang, S., & Liu, B. AI-enhanced livable environment planning: Streetscape synergies on urban thermal environment and cycling vitality in Shanghai. *Cities* (under revision).

Sun, W., Wang, L. *, **Xu, T.**, Wang, Y., Kwan, M.-P., Yang, X., Zhang, Y., Zhang, S., Chen, L., Li, Q., & Wang, C. Balancing total health gains and equity: Evaluating cooling effects of resource-constrained urban greening strategies. *Cities* (under revision).

Sun, W.†, **Xu, T.**†, Wang, L. *, Kwan, M.-P., & Wang, Y. Quantifying heat-adjusted health benefits of cycling in cities (working paper).

Xu, T., Wang, L. *, Zhao, Y., & Tao, J. (2025). Rethinking underground space utilization in metro stations: A case study of Shanghai's Jing'an Temple. 65th ACSP Annual Conference, Minneapolis, MN, USA.

Xu, T., Wang, L. *, & Zhang, S. (2025). Insights for cycle-friendly streets: Unveiling dynamic cycling route preferences. 19th IACP Conference, Xiamen, China.

Xu, T., Yang, H., Ben, B., & Zhang, Y. (2025). Reconstructing the healing system of public space: Data-driven health interventions and inclusive practices in high-density urban areas. 37th AESOP Annual Congress, Istanbul, Türkiye.

PATENT

Sun, W., Wang, L., **Xu, T.**, & Wang, Y. (2026). Method and Apparatus for Measuring Cycling Health Benefits under Comprehensive Environmental Exposure, Device, Medium, and Product. U.S. Patent Application No. 19/664,095 (pending).

SKILLS

- Python · ArcGIS · SPSS · DepthmapX · Discrete choice modeling · Map matching · Semantic segmentation
- Adobe Photoshop & Illustrator · AutoCAD